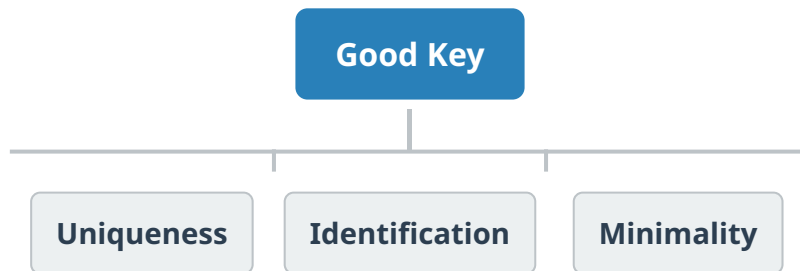


DATABASE KEYS

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Three Fundamental Properties of Keys

Every key is built around three ideas:



1. Uniqueness

A key must uniquely identify every tuple.

Example:

StudentID	Name
101	Aadya
102	Arpit
103	Aadya

- StudentID is unique.
- Name is not.

Without uniqueness, you cannot reliably identify rows.

2. Identification

A key answers:

"Which exact row are we talking about?"

Imagine a bank.

Two customers may have:

- Same name
- Same city
- Same age

But they will never have the same:

Account Number

The account number identifies the customer.

3. Minimality

This is one of the most misunderstood concepts.

A key should contain only the attributes necessary to uniquely identify a tuple.

Example:

| StudentID | Name |

Suppose StudentID alone is unique.

Then:

(StudentID)

is a key.

What about:

(StudentID, Name)

Also unique?

Yes.

But Name is unnecessary.

Therefore,

(StudentID, Name)

is not minimal.

| Types of Keys

1. Super Key

Definition

A Super Key is any set of one or more attributes that uniquely identifies each tuple in a relation.

Example

Students

StudentID	Email	Name
101	a@gmail.com	Aadya
102	b@gmail.com	Arpit

Possible Super Keys:

- StudentID
- Email
- (StudentID, Name)
- (StudentID, Email)
- (Email, Name)
- (StudentID, Email, Name)

All uniquely identify a row.

2. Candidate Key

Definition

A Candidate Key is a minimal Super Key.

Minimal means:

Remove any attribute,
and uniqueness disappears.

Example

Students

StudentID	Email	Name
101	a@gmail.com	Aadya
102	b@gmail.com	Arpit

Possible Super Keys:

- StudentID ✓
- Email ✓
- (StudentID, Name) ✗
- (Email, Name) ✗
- (StudentID, Email) ✗

Candidate Keys:

- StudentID
- Email

Because both are:

- Unique
- Minimal. Every Candidate Key is a Super Key.

But not every Super Key is a Candidate Key. candidate key is subset of super keys .

3. Primary Key

Definition

A Primary Key is the Candidate Key chosen by the database designer to uniquely identify tuples.

Characteristics

A Primary Key:

- Must be unique (only one in one table)
- Cannot be NULL
- Must remain stable
- Should be minimal .

Candidate Key	Primary Key
Potential identifier	Chosen identifier.

4. Alternate Key

Definition

Alternate Keys are Candidate Keys that were not selected as the Primary Key.

5. Composite Key

Definition

A Composite Key consists of two or more attributes used together to uniquely identify a tuple.

Example

Course Enrollment

StudentID	CourseID	Grade
101	CS101	A
101	CS102	B
102	CS101	A

Neither StudentID nor CourseID alone is unique.

Together:

(StudentID, CourseID)

is unique.

Can a Primary Key Be Composite?

Yes.

6. Natural Key

Definition

A Natural Key is a key that already exists in the real world . Examples:

- Passport Number
- Aadhaar Number.

Advantages

- Meaningful
- No extra column needed

Disadvantages

- May change
- Business rules can change
- Privacy concerns (e.g., Aadhaar)

7. Surrogate Key

Definition

A Surrogate Key is an artificial key created solely for database identification.

Example:

- StudentID = 101

The number has no business meaning.

It exists only to identify rows.

Usually implemented using:

- AUTO_INCREMENT
- IDENTITY
- SERIAL
- UUID

Advantages

- Stable
- Never changes
- Small
- Efficient for joins and indexes .

8. Foreign Key (Basic Introduction)

We'll study this in detail later.

Definition

A Foreign Key is an attribute in one table that references the Primary Key of another table.

Example:

Students

StudentID	Name
101	Aadya
102	Arpit

Orders

OrderID	StudentID
1	101
2	102

Orders.StudentID references Students.StudentID.

This establishes a relationship between the tables.

Foreign keys also help enforce referential integrity, ensuring references point to valid row

Comparisons & Concepts

Super Key vs Candidate Key

A Super Key uniquely identifies tuples but may contain unnecessary attributes.

A Candidate Key is a minimal Super Key with no redundant attributes.

Can a Candidate Key Contain NULL?

Interview Answer

No, in the relational model a Candidate Key cannot contain NULL because it must uniquely identify every tuple.

A Candidate Key must uniquely identify every tuple, so it cannot contain NULL under relational theory. In SQL, remember that UNIQUE constraints alone may permit NULL depending on the DBMS, so a true candidate key should effectively be UNIQUE and NOT NULL.